

kidneys and improve functioning of kidneys damaged by AKI. The beneficial effect of DONs was comparable to the current treatment modality, which is administration of an antioxidant drug known as N-acetylcysteine (NAC). New treatments are being sought because NAC is not easily absorbed in the kidneys.

The triangular, tubular and rectangular shapes are designed and built using a method known as DNA origami. Here, base pairing properties of DNA's four nucleotides are used to engineer and fabricate DNA origami nanostructures, which self-assemble and preferentially accumulate in kidneys.

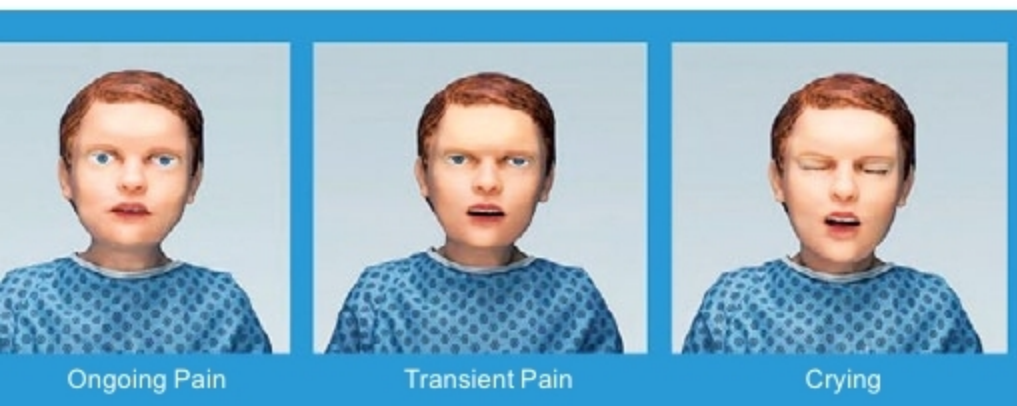
Experiments—conducted in mice as well as human embryonic kidney cells—suggest that DONs act as a rapid and active kidney protectant and may also alleviate symptoms of AKI.

Pediatric HAL, the robot child capable of bleeding, yawning and expressing pain

Developed at Gaumard Scientific, Pediatric HAL looks like a five-year-old male patient and offers unprecedented training options. The robot child is the world's most advanced pediatric patient simulator and the first capable of simulating life-like emotions through dynamic facial expressions, movement and speech. HAL is designed to help providers of all levels develop specialised skills needed to effectively communicate, diagnose and treat young patients in nearly all clinical areas.

Through scenario-based learning, HAL can help participants assess verbal and non-verbal cues to build patient-provider communication skills and empathy. In addition to illustrating nearly a dozen facial expressions, HAL also simulates a variety of common emotional states to better approximate behaviour. For example, when HAL's emotional state is set to lethargic, its eyelids droop automatically, head movement slows and yawning occurs periodically.

Interactive eyes and colour-changing skin allow Pediatric HAL to illustrate signs of varying emotional states, trauma and other neurological diseases and conditions. Owing to its ultra-high fidelity, anatomical and physiological features, the robot supports advanced algorithms using real tools and clinically-accurate techniques.



Pediatric HAL's life-like facial expressions and emotions (Credit: Gaumard Scientific)

Bionic mushrooms fuse nanotech, bacteria and fungi to make electricity

Researchers at Stevens Institute of Technology have taken an ordinary white button mushroom from a grocery store and made it bionic, supercharging it with 3D-printed clusters of cyanobacteria that generate electricity and swirls of graphene nanoribbons that can collect the current.

Manu Mannoor, assistant professor of mechanical engineering, Stevens Institute of Technology, says, "By integrating cyanobacteria that can produce electricity, with nanoscale materials capable of collecting current, we were able to better access the unique properties of both, augment them and create an entirely new functional bionic system."

Mannoor along with Sudeep Joshi, a postdoctoral fellow, showed that cyanobacterial cells lasted several days longer when placed on the cap of a white button mushroom versus a silicone and dead mushroom as suitable controls.

The duo used a robotic arm-based 3D printer to first print an electronic ink containing graphene nanoribbons. This printed branched network served as an electricity-collecting network atop the mushroom's cap by acting like a nano-probe, to access bio-electrons generated inside cyanobacterial cells.

Next, they printed a bio-ink containing cyanobacteria onto the mushroom's cap in a spiral pattern intersecting with the electronic ink at multiple contact points. At these locations, electrons could transfer through the outer membranes of the cyanobacteria to the conductive network of graphene nanoribbons. Shining a light on the mushrooms activated cyanobacterial photosynthesis, generating a photocurrent.



A white button mushroom equipped with 3D-printed graphene nanoribbons (black), which collects electricity generated by densely-packed 3D-printed cyanobacteria (green) (Credit: Stevens Institute of Technology)